Tribhuvan University

Faculty of Science and Technology

OFFICE OF THE DEAN

2080

Bachelor of Science in Computer Science and Information Technology

Course Title: **Operating Systems**

Code No: **CSC264** Semester: **4th Semester**

Full Marks: **60** Pass Marks: **24** Time: **using FIFO, SSTF and SCAN. Assume that the initial position of head is at 100.2.How do you distinguish between deadlock and starvation? Describe the necessary conditions for deadlock. Explain the working mechanism of TLB.3.Why do we need to schedule process? Find the average waiting time and average turnaround time for the following set of processes using FCFS, SJF, RR(Quantum: 3) and Shortest Remaining Time Next.ProcessCPU Burst TimeArrival TimeP1200P22515P31030P41545Section BAttempt any Eight questions[8×5=40]4.What is system call? Describe the transition between different states of a process.5.Discuss about contiguous and linked list allocation approach in implementing files.6.Why do we need virtual memory? Describe the structure of a page table.7.Illustrate the term safe and unsafe state in deadlock prevention with a scenario.8.How lock variable is used in achieving mutual exclusion? Describe.9.Why do we need hierarchical directory system? Explain the structure of a disk.10.Find the number of page faults using FIFO and LRU for the reference string, 4, 3, 6, 1, 7, 6, 1, 2, 1, 2 with frame size 3.11.Define working set. How does clock page replacement algorithm work?12.Write short notes ona. Inodesb. RAID**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. How DMA operation is performed? Consider a disk with 200 tracks and the queue has random requests from different processes in the order: 45, 48, 29, 17, 80, 150, 28, and 188. Find the seek time using FIFO, SSTF and SCAN. Assume that the initial position of head is at 100.

2. How do you distinguish between deadlock and starvation? Describe the necessary conditions for deadlock. Explain the working mechanism of TLB.

3. Why do we need to schedule process? Find the average waiting time and average turnaround time for the following set of processes using FCFS, SJF, RR(Quantum: 3) and Shortest Remaining Time Next.ProcessCPU Burst TimeArrival TimeP1200P22515P31030P41545

Section B

Attempt any Eight questions[8×5=40]

4. What is system call? Describe the transition between different states of a process.

5. Discuss about contiguous and linked list allocation approach in implementing files.

6. Why do we need virtual memory? Describe the structure of a page table.

7. Illustrate the term safe and unsafe state in deadlock prevention with a scenario.

8. How lock variable is used in achieving mutual exclusion? Describe.

9. Why do we need hierarchical directory system? Explain the structure of a disk.

10. Find the number of page faults using FIFO and LRU for the reference string, 4, 3, 6, 1, 7, 6, 1, 2, 1, 2 with frame size 3.

11. Define working set. How does clock page replacement algorithm work?

12. Write short notes ona. Inodesb. RAID